

AstroMesh — demo script

Round 2 · Challenge #518 — Constraint Awareness: Completion Signal Three minutes. Spoken text in plain type, actions in **bold**.

1 · What it is · 0:00–0:25

[Boot, globe flies in]

AstroMesh is a collision-avoidance system for satellites.

In Round 1 we built the operational layer: a live catalogue of thirty-one thousand tracked objects, all-pairs conjunction screening, an avoidance planner that checks a new orbit against the whole catalogue, launch-window planning, multi-operator consensus voting, and a natural-language mission-control assistant.

That system told you about risk. Round 2 makes it refuse to act on risk it cannot justify.

2 · The capability · 0:25–0:45

[Point at the signal card]

We added a constraint gate. Every irreversible action passes through it, and it reports one of four states: complete, partial, blocked, or unresolved.

Two rules hold everywhere. Unresolved is never treated as a pass. And you cannot waive a rule that was never measured.

Right now the fleet signal reads **BLOCKED**, so nothing can be authorised.

3 · A refusal, with reasons · 0:45–1:15

[RISK tab → VOTE on the top card]

Here is a real conjunction. I'll put it to the operator vote.

The gate refused it and listed why. Flight Rule seven requires that at least one of the two objects can be commanded. Both are debris, so neither can be moved. That rule is non-negotiable — there is no waiver path in the engine, so a unanimous vote cannot approve it.

The rows in grey are rules we could not evaluate at all. We report those as unresolved and treat them as blocking, because not knowing is not the same as being fine.

4 · Where a verdict comes from · 1:15–2:00

[HOW tab — corridor draws on the globe]

This walks one real event through every layer: the Long March 5B stage that re-entered in November 2022 and closed airspace over Spain and France.

Each step gives a number and its source.

Corridor width comes from the live geomagnetic index — density uncertainty of plus or minus forty-nine percent, measured from our own calibration.

The ground underneath is classified by DINOv3, a satellite vision model, across eighteen million cells at three-point-four kilometres. We validated it against population data it had never seen. Above sixty degrees north it stops classifying, because we measured that it becomes unreliable there.

Casualty expectancy for this footprint is twenty times the legal limit.

And this step is the finding. The standard regulatory method averages population into latitude bands, so it cannot tell which longitude the debris is heading for. On the same footprint it reports half the risk. Across eighteen thousand footprints, it understates exposure ninety-four percent of the time over densely populated ground.

[FOLLOW DESCENT — 10 seconds]

The same descent from the vehicle. Altitude, flight path angle, downrange. The corridor is the recorded event; the motion is a replay, and the caption says so.

5 · Constraints move · 2:00–2:25

[STORMS tab → REPLAY GANNON]

This is the Gannon storm of May 2024, when about half of all low-orbit satellites manoeuvred within days.

Radio fails over the poles, navigation degrades in the auroral regions, and our reachable ground stations drop from eight to three. Flight Rule nineteen needs the command uplink available long enough to send the manoeuvre, so the deadline to decide moves earlier.

A constraint is not a fixed checkbox. Real conditions can remove your ability to act while you are still deciding.

6 · Independent of the theme · 2:25–2:45

[PROVE IT]

The challenge required this to work outside any one domain.

Same engine, no code changes, running a software release gate instead — tests, security vulnerabilities, rollback plans, error budgets. Same four states, same precedence. We also encode an FAA aircraft-dispatch rulebook the same way.

The rules are data. Orbital operations is just where we needed it first.

7 · Close · 2:45–3:00

[FILTERS → GPU SCREEN, then back to the signal]

This screened every pair in the catalogue, about half a billion, on the laptop's GPU. It is a coarse filter — candidates are re-screened with full propagation before anything acts.

Four states: complete, partial, blocked, unresolved. The repository has the supporting work: a model checker over a hundred and forty-eight thousand states, property tests across generated rulebooks, and coverage guarantees for the statistical bounds.

Two of those results describe the limits of what the system can determine.

Round 2 — the checklist

Say these if there is time; each is on screen.

Point	Where
Four states with strict precedence, blocked over unresolved	signal card
Non-negotiable rules with no waiver path	FR-05, FR-07, FR-08 in the rulebook
Unresolved as a blocking state	grey rows
Two-tier evidence: operator ephemeris 3 m vs TLE 1,689 m	FR-20
Live official re-entry predictions	HOW, step 4
Vision model with a measured refusal band	HOW, step 5
2-D exposure vs the 1-D regulatory method	HOW, step 9
Countdown to self-blocking	signal card, "self-blocks in"
Cheapest set of measurements that would clear it	HOW, unresolved steps
Decision receipts, hash-chained and replayable	HOW, bottom
Quorum loss blocks everything	FLEET → STOP two nodes
Autonomous agent bound by the same gate	MISSION → ACTIVATE

Notes

- One take per scene; cut between them.
- Don't narrate clicking. Click, then speak to what appears.
- When the gate refuses, don't apologise — that is the product working.
- Never say "formally verified" without saying what was checked and to what bounds.
- Keep the recorded fallback cued.